

Multiple Spatial–Spectral Features Aggregated Neural Network for Hyperspectral Change Detection

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Abstract—Recently, convolution neural networks (CNNs) have flourished in hyperspectral image (HSI) change detection (CD). However, these approaches typically rely on single-scale and single-level features to obtain change information, limiting the further boost detection accuracy. To solve the above problem, we propose a novel multiscale and multilevel spatial–spectral features aggregated neural network (M²S²Net) for HSI CD. The proposed method leverages the multiple spatial–spectral (SS) features of bi-temporal images to mine temporal change information. First, the HSIs are cropped into patches with different spatial and spectral sizes to provide various scales SS information and alleviate the problem of spectral redundancy. Then, the patches are utilized to capture the multiscale SS features in the residual connection block (ResBlock). These bi-temporal features are inputted into the Transformer encoder-based feature fusion module to learn the discriminative change representations. Via the global receive field of the self-attention mechanism, various fine-grained change information is learned from the features at different scales. Finally, the change features of each level are aggregated to produce the change map. The experiments on two public HSI datasets verify the effectiveness of the proposed method. Specifically, the overall accuracy (OA) of the M²S²Net surpasses the second-best method by 0.56% and 0.10% on the Farmland and River datasets, respectively.

Index Terms—Attention mechanism, change detection (CD), hyperspectral image (HSI), multiscale features.

I. INTRODUCTION

CHANGE detection (CD) involves the analysis of images captured at different time intervals to identify and quantify changes in a specific area. This technique finds extensive application in land use, disaster monitoring, and agricultural monitoring. The utilization of hyperspectral images (HSIs), which consist of numerous continuous bands, offers a higher level of precision in analyzing ground objects. Consequently, HSI CD has garnered significant interest and attention.

Traditional approaches, such as change vector analysis (CVA) [1], slow feature analysis (SFA), and tensor decomposition and reconstruction detector (TDRD) [2], exploit change information through simple mathematical operations. Although they have gained some achievements, they are challenging to exact discriminative semantic features and are limited to further performance improvement.

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Over the past few years, HSI CD has witnessed the integration of deep learning (DL) techniques, such as autoencoders (AE), convolutional neural networks (CNNs), and recurrent neural networks (RNNs). With its excellent spatial feature capture and representation capabilities, CNNs are the most widely used DL-based method for HSI CD. Wang et al. [3] used a 2-D CNN to learn distinguishing patterns from a mixed-affinity matrix. Song et al. [4] achieved effective HSI CD by separately optimizing spectral and spatial feature extraction before fusion. Attention mechanisms allow models to focus on the most relevant parts of a large input, enabling more efficient and accurate data modeling. Gong et al. [5] employed adaptive spectral and spatial attention (SA) mechanisms to suppress irrelevant information and enhance relevant features for CD. The self-attention mechanism [6] can learn the correlation between various positions in the input sequence, and has achieved results in remote sensing. Wang et al. [7] utilized spectral–spatial–temporal transformers (SST-Formers) to model spectral and spatial sequences and capture time dependencies between images.

While these methods have yielded noteworthy CD results, there remains room for continued improvement. The changed regions within diverse scenes often exhibit varying physical sizes, yet the above methods employ single-sized image patches as input. This makes it difficult to effectively detect both substantial-scale and subtle changes. Additionally, these methods use the Siamese-based network to learn respective semantic information of the bi-temporal HSIs, and then use them to extract temporal change dependency. This leads to the ignorance of temporal information during the feature extraction of each image.

Several approaches have endeavored to capture feature representations at various scales by employing convolutional kernels of different sizes. Yang et al. [8] utilized multiscale 3-D convolution to extract SS features across multiple scales. However, these features are simply concatenated, leading to feature confusion and redundancy. Moreover, the adoption of 3-D convolution brings an increase in computational complexity. Ou et al. [9] used a multiscale spatial attention mechanism to assign the highest weight to the central pixel but overlooked the influence of spectral information across diverse scales. The numerous bands of HSIs can not only offer fine-grained ground object information but also bring more noise and information redundancy. Therefore, effective measures should be taken to mitigate spectral redundancy.

Based on the above analysis, we propose a novel neural network based on multiscale and multilevel SS features

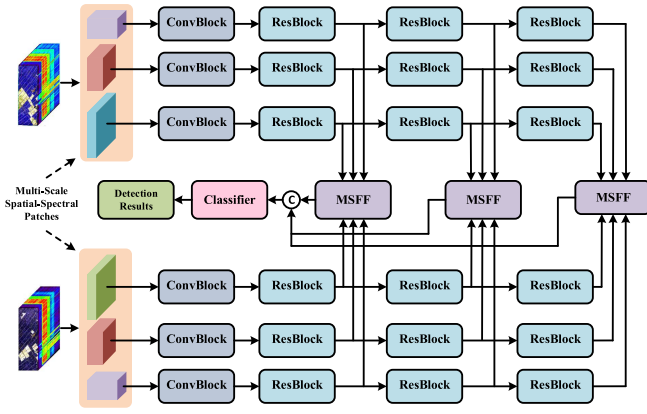


Fig. 1. Overall structure of the proposed method. First, the HSIs are cropped into multiscale SS patches and used to extract multiscale features. Next, the features are aggregated by the MSFF module to capture temporal change information at each level. Finally, the aggregated change information of each MSFF module is leveraged to produce the change map.

(M²S²Net). The bi-temporal HSIs are cropped into image patches with various spatial and spectral sizes to capture multiscale SS features. The image patch with large spatial and small spectral size provides large-scale spatial information while reducing the noise and redundancy of spectral information. The image patch with small spatial and large spectral size improves the model's ability to detect subtle changes and enables the model to leverage the rich spectral information. Next, the multiscale features of each level are aggregated in the multiscale features fusion (MSFF) module based on the Transformer (Trm) encoder. The self-attention mechanism enables global modeling ability, capturing temporal change information with various fine-grained granularity from the bi-temporal multiscale SS features. Finally, the multilevel change features learned at each MSFF module are leveraged to produce the change map. In this letter, the main contributions are highlighted as follows.

- 1) We propose a novel M²S²Net for HSI CD, which aggregates the temporal change information mined from multiscale SS features at each layer to improve the detection performance.
- 2) We build multiscale image patches with different spatial and spectral sizes around the central pixel to mine multiscale SS features and alleviate spectral redundancy.
- 3) The Transformer is utilized to capture temporal dependency from multiscale SS features for the first time. The proposed MSFF module based on Transformer encoders effectively learns the importance and intrinsic relevance of different scale features to obtain a more discriminative change feature.

The subsequent sections are structured as follows: Section II details the M²S²Net framework. Section III validates the performance of our proposed method. Finally, Section IV concludes this letter.

II. PROPOSED METHOD

This section introduces our proposed M²S²Net. The overall structure of M²S²Net is shown in Fig. 1.

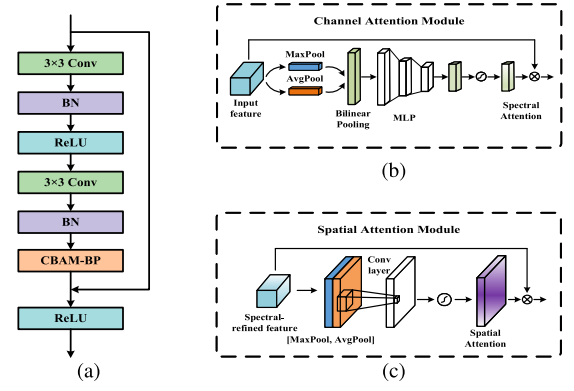


Fig. 2. Structure. (a) ResBlock. (b) CA. (c) SA.

A. Multiscale SS Patch Crop

The HSIs are cropped into multiscale patches to learn SS features at various scales. For the HSI $I_t \in \mathbb{R}^{B \times H \times W}$ ($t = 1, 2$) with the spatial size of $H \times W$ and band number of B , multiscale image patches $X_{t,s} \in \mathbb{R}^{(B/2^{s-1}) \times (2s+1) \times (2s+1)}$ ($s = 1, 2, \dots, S$) are cropped from image I_t . $X_{t,s}$ is generated by randomly selecting one band for every 2^{s-1} bands, mitigating the problem of spectral redundancy.

Features at different scales can capture different details and structures and reduce noise interference at a single scale, improving the accuracy and robustness of the model.

B. Multiscale SS Feature Extraction

A Siamese-based network containing convolutional block (ConvBlock) and residual connection block (ResBlock) is utilized to acquire the multiscale SS features from the patches at various scales.

First, the ConvBlock is used to capture the primary features from $X_{t,s}$

$$F_{t,s}^0 = \text{ReLU}(\text{BN}(\text{Conv}(X_{t,s}))) \quad (1)$$

where $F_{t,s}^0 \in \mathbb{R}^{C \times (2s+1) \times (2s+1)}$ denotes the primary feature with the dimension of C , ReLU denotes the rectified linear unit, BN is the batch normalization layer, and Conv refers to the 2-D convolution layer with kernel size equal to the patch size. Next, the ResBlock is used to extract SS features of each scale. Fig. 2 shows the detailed structure of ResBlock. In ResBlock, the features $F_{t,s}^0$ are first used to extract coarse features $F_{t,s}$

$$F_{t,s} = \text{BN}(\text{Conv}(\text{ReLU}(\text{BN}(\text{Conv}(F_{t,s}^0))))). \quad (2)$$

To improve the network's capacity to represent image features, the convolutional block attention module with bilinear pooling (CBAM-BP) is integrated within the ResBlock. As shown in Fig. 2(b) and (c), CBAM-BP consists of channel attention (CA) and SA, which enhance important information and suppress useless information in the spectral and spatial domains, respectively. The overall procedure of CBAM-BP can be formulated as

$$F'_{t,s} = M_{ca}(F_{t,s}) \otimes F_{t,s} \quad (3)$$

$$F''_{t,s} = M_{sa}(F'_{t,s}) \otimes F'_{t,s} \quad (4)$$

where $\otimes$ denotes element-wise multiplication, $F''_{t,s}$ is the final attention-enhanced feature. $M_{ca}(F_{t,s})$ and $M_{sa}(F'_{t,s})$ are the

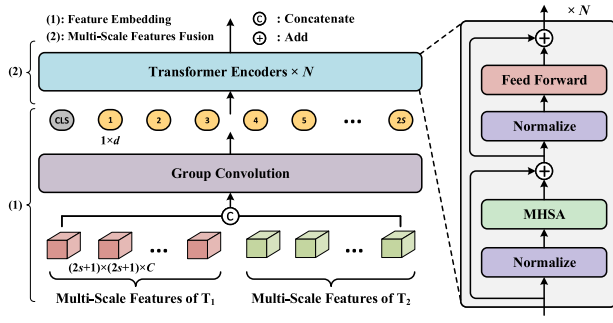


Fig. 3. Illustration of MSFF module. The multiscale features of HSIs are concatenated and converted into feature embeddings to aggregate the temporal change information by the Transformer encoders.

channel and SA maps, respectively. $M_{ca}(F_{t,s})$ and $M_{sa}(F'_{t,s})$ are computed as

$$M_{ca}(F_{t,s}) = \sigma(\text{MLP}(\text{BP}(\text{AvgPool}(F_{t,s}), \text{MaxPool}(F_{t,s})))) \quad (5)$$

$$M_{sa}(F'_{t,s}) = \sigma(\text{Conv}([\text{AvgPool}(F'_{t,s}); \text{MaxPool}(F'_{t,s})])) \quad (6)$$

where σ denotes the Sigmoid activation function, MLP is a multilayer perceptron, BP denotes the bilinear pooling [10], and its calculation process can be presented as

$$\phi(F_{t,s}) = \text{vec}(\text{AvgPool}(F_{t,s})^T \text{MaxPool}(F_{t,s})). \quad (7)$$

After the bilinear feature vector $\phi(F_{t,s}) \in \mathbb{R}^{C^2 \times 1}$ is processed by the signed square-root step ($u_t \leftarrow \text{sign}(\phi)\sqrt{|\phi|}$) and ℓ_2 normalization ($z_t \leftarrow u_t / \|u_t\|_2$), it is used for the following attention computation.

Finally, the refined features $F''_{t,s}$ are integrated with the initial input $F_{t,s}$ via a residual connection

$$F''_{t,s} = \text{ReLU}(F_{t,s}^{i-1} + F_{t,s}^{i-1}), \quad i = 1, 2, 3 \quad (8)$$

where $F''_{t,s}$ denotes the feature obtained by the i th ResBlock of image patch $X_{t,s}$. By constructing multiscale image patches with inconsistent spatial and spectral sizes around the central pixel, multiscale SS features are captured, and redundant information is removed. Multiscale spatial features enable the model to effectively detect subtle changes and identify large-scale change areas. Multiscale spectral features alleviate the problem of spectral redundancy while leveraging the rich spectral information.

C. Multiscale SS Features Fusion

As shown in Fig. 3, the features $F''_{t,s}$ are fed into the MSFF module at each level to aggregate the SS features with different scales.

1) *Feature Embedding*: We employ group convolution to convert the multiscale features into feature embeddings, which leverages the multiscale spatial information within each channel. The convolution kernel size is equal to the size of the feature map, and no padding is needed

$$E = [\text{CLS}; \text{Concat}_{t=1}^2 (\text{Concat}_{s=1}^S (\text{GConv}(F''_{t,s})))] \quad (9)$$

where $E \in \mathbb{R}^{(2S+1) \times d}$ is the feature embedding sequence with the dimension of d , GConv denotes the group convolution with the group size of C and the kernel size of $(2s+1) \times (2s+1)$, and $\text{CLS} \in \mathbb{R}^{1 \times d}$ is the randomly initialized class token used to gather change information from various scales of features.

2) *Multiscale Features Fusion*: Transformer uses self-attention to learn relationships between all positions in a sequence regardless of their positions, allowing it to capture intricate long-range dependencies. Therefore, the MSFF module based on Transformer encoders is introduced to aggregate multiscale SS features and generate more discriminative change features. The multihead self-attention (MHSA) mechanism, a feed-forward network (FFN) layer, layer normalization (LN), and residual connection are included in one Transformer encoder. Specifically, the output X_l of the l th Transformer layer ($l \in [1, 2, \dots, L]$) can be represented as

$$X'_l = \text{MHSA}(\text{LN}(X_{l-1})) + X_{l-1} \quad (10)$$

$$X_l = \text{FFN}(\text{LN}(X'_l)) + X'_l \quad (11)$$

where MHSA can be formulated as

$$\text{MHSA}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (12)$$

Our proposed MSFF module leverages multiscale SS features from the CNN to achieve an effective fusion of multigranularity features through the Transformer's MHSA mechanism. Specifically, the features with small spatial sizes contain more rich spectral information and enable the model to detect subtle changes. The features with large spatial sizes provide large-scale spatial information and reduce the noise and redundancy of the spectrum. Compared with simple concatenation, the SA mechanism learns the weights of different features for more effective feature fusion. Therefore, the fused features not only contain rich SS information but also learn the intrinsic correlation between the features, yielding more discriminative change representations.

D. Prediction and Loss Function

The CLS tokens from the final Transformer encoder layer of each MSFF module are concatenated and provided as input to the classification head. The classification head constructed by a fully connected network and softmax function produces the detection result.

The CD task can be regarded as a binary classification problem, so the binary cross-entropy (BCE) is adopted as the loss function to optimize the M²S²Net.

III. EXPERIMENT RESULTS AND ANALYSIS

A. Datasets

Two datasets are utilized to validate the effectiveness of our proposed approach. The first dataset, Farmland,¹ covers an area of Farmland in Yancheng City, China. The images were obtained on May 3, 2006, and April 23, 2007. Each image has a spatial size of 420×140 , and composes 154 spectral

¹Available at rslab.ut.ac.ir.

TABLE I
OPTIMAL PARAMETERS FOR DIFFERENT DATASETS

	epoch	batch size	scale (S)	Trm layers	Trm heads	dim
Farmland	100	64	2	1	8	128
River	50	32	3	4	4	128

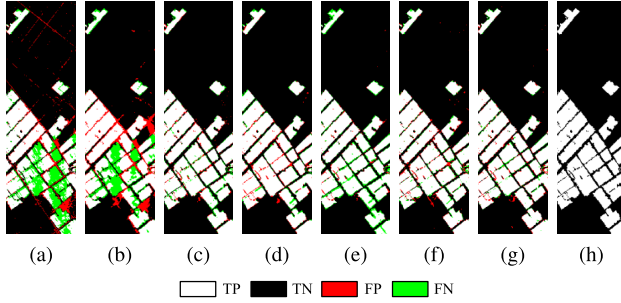


Fig. 4. Detection results for Farmland. (a) CVA. (b) TDRD. (c) Re3FCN. (d) CSANet. (e) SST-Former. (f) ML-EDAN. (g) Proposed. (h) Ground Truth.

TABLE II
QUANTITATIVE COMPARISON ON FARMLAND DATASETS

	CVA	TDRD	Re3FCN	CSANet	SST-Former	ML-EDAN	Proposed
OA	0.8749	0.8847	0.9623	0.9620	0.9507	0.9622	0.9678
Kappa	0.6998	0.7309	0.9122	0.9118	0.8856	0.9121	0.9252

bands. The second dataset, River,² was collected on May 3, 2013, and December 31, 2013, respectively. The size of the images is 463×241 with 198 spectral bands. The main change is the substance of a river in Jiangsu Province, China.

B. Experimental Setup

To demonstrate the superiority of our proposed M²S²Net, we select the following comparison methods: CVA [1], TDRD [2], recurrent 3-D fully convolutional network (Re3FCN) [11], cross-temporal interaction symmetric attention network (CSANet) [12], SST-Former [7], and multilevel encoder-decoder attention network (ML-EDAN) [13]. Moreover, overall accuracy (OA) and the Kappa coefficient are employed to verify the detection performance.

We implement the proposed M²S²Net using Pytorch on a single GTX 1080Ti GPU. The optimizer is Adam and the learning rate is $5e-4$. In the experiments, the ratio of both training samples and validation samples is set to 5%. The other optimal parameters for each dataset are shown in Table I.

C. Experimental Results

1) *Farmland*: The visual and quantitative performance is shown in Fig. 4 and Table II, respectively. The edges of the changed regions are blurred and even adhered in the change maps of CVA and TDRD, and there are also massive miss detections in green in the lower middle part of the image. So, these two methods also achieve the worst OA and Kappa. For Re3FCN, some miss and false detections are still presented in

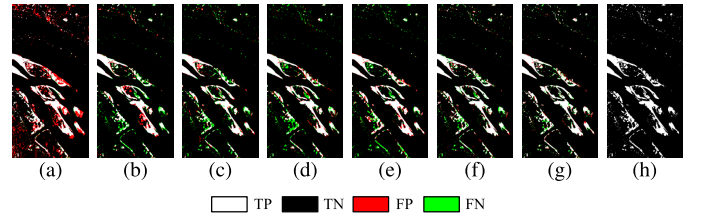


Fig. 5. Detection results for River. (a) CVA. (b) TDRD. (c) Re3FCN. (d) CSANet. (e) SST-Former. (f) ML-EDAN. (g) Proposed. (h) Ground Truth.

TABLE III
QUANTITATIVE COMPARISON ON RIVER DATASETS

	CVA	TDRD	Re3FCN	CSANet	SST-Former	ML-EDAN	Proposed
OA	0.9267	0.9615	0.9626	0.9623	0.9632	0.9704	0.9714
Kappa	0.6575	0.7475	0.7485	0.7411	0.7460	0.8037	0.8161

the edge of the changed areas, though in a reduced number than unsupervised methods. Compared to Re3FCN, CSANet exhibits a higher count of false detections in red than miss detections in green, but their OA and Kappa are almost the same. The change map of SSF-Former demonstrates a higher count of miss detections, particularly at the edge of the change region in the upper left corner, which is consistent with its OA and Kappa being lower than the other supervised methods. ML-EDAN and our proposed method exhibit comparable and superior visual performance. However, a closer inspection reveals that ML-EDAN displays a higher count of green miss detections in the lower right area of the image. Correspondingly, our M²S²Net achieves the highest OA and Kappa among all the methods, measuring 96.78% and 0.9252%, respectively. The proposed method's superior performance is attributed to the utilization of multiscale and multilevel SS change features.

2) *River*: Fig. 5 and Table III illustrate the detection results on the River dataset. It can be seen that CVA obtains the worst results. There are numerous red false detection areas in the change map, and its OA and Kappa are only 92.67% and 0.6575%, respectively. TDRD is an unsupervised method, but its OA and Kappa are close to the supervised approaches. However, TDRD exhibits poor performance on the other dataset, indicating a lack of robustness when applied to datasets with different scenes. Although the OA and visual performance of Re3FCN, SST-Former, and CSANet are similar, the Kappa of Re3FCN and SST-Former are better than the latter, approximately 0.049 higher than CSANet. The lower Kappa suggests that the method's predictions are in poor consistency with the actual detection results. ML-EDAN has achieved the second-best performance among all comparison methods and is the only one with an OA above 97%. This highlights the effectiveness of multilevel features in enhancing the accuracy of CD. Our proposed method attains the highest detection results with an OA of 97.14% and a Kappa of 0.8161%. Although the OA of our proposed method is slightly better than ML-EDAN, the Kappa shows a significant increase of 0.0124. This improvement can be attributed to our proposed method's ability to better learn change rules from multiscale SS features.

²Available at <http://crabwq.github.io>.

TABLE IV
ABLATION STUDY OF EACH MODULE ON RIVER DATASET

Model	(1)	(2)	(3)	(4)	(5)	(6)
MSFF	×	✓	✓	✓	×	✓
Multi-Scale Features	×	×	✓	×	✓	✓
Multi-Level Features	×	×	×	✓	✓	✓
OA	0.8628	0.9668	0.9665	0.9703	0.7788	0.9714
Kappa	0.4796	0.7838	0.7937	0.8104	0.3458	0.8161

TABLE V
RESULTS OF DIFFERENT SCALES ON TWO DATASETS

Dataset	Metric	Scale (S)						
		1:(3)	1:(5)	1:(7)	1:(9)	2:(3,5)	3:(3,5,7)	4:(3,5,7,9)
Farmland	OA	0.9631	0.9655	0.9656	0.9639	0.9678	0.9655	0.9662
	Kappa	0.9147	0.9200	0.9205	0.9164	0.9252	0.9200	0.9214
River	OA	0.9700	0.9704	0.9703	0.9672	0.9700	0.9714	0.9694
	Kappa	0.8090	0.8054	0.8104	0.7940	0.8101	0.8161	0.8107

TABLE VI
AVERAGE COMPUTATION TIME OF DIFFERENT ALGORITHMS
ON RIVER DATASET (UNIT: SECONDS)

	CVA	TDRD	Re3FCN	CSANet	SST-Former	ML-EDAN	Baseline	Proposed
Training	—	—	16.34	337.7	755.67	1565.07	505.09	769.25
Inference	0.18	20.13	2.05	13.6	11.38	27.22	17.31	20.04

D. Ablation Study

In our work, the multiscale patches, multilevel features, and the MSFF module play critical roles. To evaluate their effectiveness, we conducted a series of ablation experiments.

1) *MSFF*: From Table IV, we can see that the inclusion of MSFF in Model (2) results in a remarkable performance than Model (1) without MSFF. Moreover, the results of Model (6) outperform Model (5) after the addition of the MSFF module. This indicates that our proposed MSFF can enhance the leverage of temporal change information.

2) *Multiscale Features*: As shown in Table IV, the performance of Model (2) over Model (1) and Model (6) over Model (4) are all improved, which shows that multiscale SS features can improve the performance of CD.

3) *Multilevel Features*: It can be seen from Table IV that the integration of multilevel features leads to an improvement in performance for Model (6) compared to Model (3). This is because the multilevel features provide the low-level details and high-level semantics.

E. Discussion

1) *Multiscale Features*: We conducted experiments to analyze the effect of different scales of multiscale SS features on the model performance. Table V shows that we can observe that the two datasets achieve the best performance at scales of 2:(3, 5) and 3:(3, 5, 7), respectively. Notably, 2:(3, 5) means

the image is cropped into two patches with a spatial size of three and five, respectively. When the scale is small, it provides too little spatial and spectral information; when it is too large, it brings much redundant information and noise.

2) *Time Cost*: As shown in Table VI, we compare the time cost of different methods on the River dataset. Notably, the case with a single-scale feature is denoted as Baseline. Our proposed M²S²Net has the second most training and inference time, but the training time is significantly less than ML-EDAN.

IV. CONCLUSION

In this letter, we propose a novel multiscale and multilevel SS features aggregated neural network for HSI CD, where the HSIs are cropped into patches with different scales and used to capture multiscale SS features and alleviate spectral redundancy, and the Transformer encoder-based MSFF module is introduced to learn the various fine-grained change information. The change features at each level are aggregated to obtain the CD results. The experiments on two datasets demonstrate that our proposed approach surpasses the current methods.

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